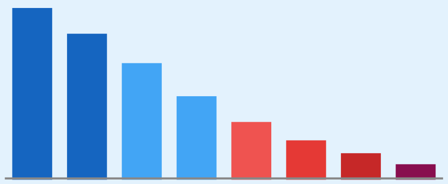
1 PROBLEM

Long-Tail Distribution Problem



Head classes: 5,000+ imgs

Seen vs unseen category imbalance ⚠

LVIS v1.0 Dataset

1,203 Categories	337 Long-tail categories	164K Images

COCO-ZSD Benchmark

48 Seen | 17 Unseen ★

Limitations of Existing OVD

- ✕ Single prompt — no diversity
- ✕ No uncertainty modelling
- ✕ Unseen categories remain challenging under generalized zero-shot recognition

2 METHOD

Semantic Prompt Generation

WordNet: Synonyms · Hyponyms · Hypernyms

LLM Paraphrase: GPT diverse descriptions

Prompt Pool (1-10 per class)

armadillo · nine-banded · armored mammal · dasypod · cingulate

CLIP ViT-L/14 Encoding

Image Encoder 768-d — Text Encoder 768-d

Cosine Similarity → Score Matrix (N × M)

Preprocessing Pipeline:

- Resize 640×640 · Normalise
- Parse annotated object regions
- LVIS + COCO-ZSD semantic evaluation setup

3 FUSION CORE

★ Semantic Fusion Strategies

Mean Fusion
Arithmetic mean of all prompt similarity scores — AP_unseen: 67.5%

Max Fusion
Highest similarity score across prompt set — AP_unseen: 63.1%

Weighted Fusion
Softmax-based weighting — AP_unseen: 67.7%

Semantic Evidence Fusion

- Aggregates complementary semantic prompt evidence
- Entropy-guided uncertainty estimation
- Frequency-aware semantic weighting

Entropy-aware fusion provides diagnostic calibration signals

4 RESULTS

Key Quantitative Results

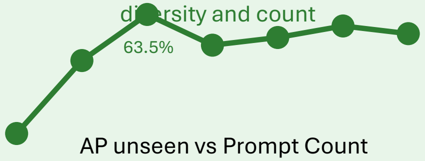
Weighted Fusion achieves highest GZSD HM

AP_unseen
63.5% → 67.7%
(+4.2 pts)

GZSD HM
58.4% → 61.2%
(+2.8 pts)

Stable robustness across prompt perturbations
vs 0.76 baseline

Performance varies with prompt diversity and count



AP unseen vs Prompt Count

Improved semantic uncertainty estimation

5 IMPACT

Research Contributions

- **RQ1**: Prompt fusion as evidence +6.1 AP_rare
- **RQ2**: Weighted fusion provides the most balanced GZSD performance
- **RQ3**: Entropy-aware fusion provides semantic uncertainty estimation
- **RQ4**: Prompt diversity improves semantic robustness
- **RQ5**: Frequency-aware weighting (\alpha = 0.4)
- **RQ6**: Stable robustness across prompt perturbations

Scalable · No extra training data
Info Fusion + CV community impact